

● HARD

● ALGEBRA

● ANSWER KEY

Algebra

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01

Q1**1.2**

Accept also: 6/5

The correct answer is $\frac{6}{5}$. It's given that line j is perpendicular to line k in the xy -plane. This means that the slope of line j is the opposite reciprocal of the slope of line k . For a line that passes through the points x_1, y_1 and x_2, y_2 in the xy -plane, the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. It's shown that line k passes through the points $(-6, 0)$ and $(0, -5)$ in the xy -plane. Substituting -6 for x_1 , 0 for y_1 , 0 for x_2 , and -5 for y_2 in $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{-5 - 0}{0 - (-6)}$, or $-\frac{5}{6}$. The opposite reciprocal of $-\frac{5}{6}$ is $\frac{6}{5}$. Therefore, the slope of line j is $\frac{6}{5}$. Note that $6/5$ and 1.2 are examples of ways to enter a correct answer.

Q2**.0465**

Accept also: 2/43

The correct answer is $\frac{2}{43}$. It's given that $f(cx) = x - 8$ for all values of x , where c is a positive constant, and $f(2) = 35$. Therefore, for the given function f , $cx = 2$. Dividing both sides of this equation by c yields $x = \frac{2}{c}$. Substituting $\frac{2}{c}$ for x in the equation $f(cx) = x - 8$ yields $f\left(\frac{2c}{c}\right) = \frac{2}{c} - 8$, or $f(2) = \frac{2}{c} - 8$. Since it's given that $f(2) = 35$, substituting 35 for $f(2)$ yields $35 = \frac{2}{c} - 8$. Adding 8 to both sides of this equation yields $43 = \frac{2}{c}$. Multiplying both sides of this equation by c yields $43c = 2$. Dividing both sides of this equation by 43 yields $c = \frac{2}{43}$. Note that $2/43$, $.0465$, 0.046 , and 0.047 are examples of ways to enter a correct answer.

Q3

B

B. $140x + 3 + 220x \geq 5,820$

Choice B is correct. It's given that a team plans to sell at least 140 tickets before an event and at least 220 tickets during the event to raise a total of at least \$ 5,820 from all tickets sold. It's also given that the price of a ticket during the event is \$ 3 less than the price of a ticket before the event and that x represents the price, in dollars, of a ticket sold during the event. It follows that $x + 3$ represents the price, in dollars, of a ticket sold before the event. The expression $140(x + 3)$ represents the planned revenue, in dollars, from the tickets sold before the event, and the expression $220x$ represents the planned revenue, in dollars, from the tickets sold during the event. Thus, the expression $140(x + 3) + 220x$ represents the planned revenue, in dollars, from all tickets sold. Since the team plans to raise a total of at least \$ 5,820 from all tickets sold, the total revenue must be at least \$ 5,820. Therefore, the inequality $140(x + 3) + 220x \geq 5,820$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which the team raises a total of at most \$ 5,820 from all tickets sold.

Choice C is incorrect. This inequality represents a situation in which the price of a ticket before the event is \$ 3 less, rather than \$ 3 more, than the price of a ticket during the event and the team raises a total of at most \$ 5,820 from all tickets sold.

Choice D is incorrect. This inequality represents a situation in which the price of a ticket before the event is \$ 3 less, rather than \$ 3 more, than the price of a ticket during the event.

Q4

B

B. -3 and 3

Choice B is correct. Multiplying both sides of the given equation by $(7)(9)$, or 63, yields $\left(63\right)\frac{x-5}{7} = \left(63\right)\frac{x-5}{9}$, or $9(x-5) = 7(x-5)$. Subtracting $7(x-5)$ from both sides of this equation yields $2(x-5) = 0$. Dividing both sides of this equation by 2 yields $x-5 = 0$. Therefore, if $\frac{x-5}{7} = \frac{x-5}{9}$, then the value of $x-5$ is 0. It follows that of the given choices, the value of $x-5$ is between -3 and 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

3.5

Accept also: 7/2

The correct answer is $\frac{7}{2}$. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are distinct and parallel. Two lines represented by equations in standard form $Ax + By = C$, where A , B , and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system, $\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$, can be written in standard form by adding $\frac{7}{8}y$ to both sides of the equation, which yields $\frac{14}{8}y - \frac{5}{8}x = \frac{4}{7}$, or $-\frac{5}{8}x + \frac{14}{8}y = \frac{4}{7}$. Multiplying each term in this equation by -8 yields $5x - 14y = -\frac{32}{7}$. The second equation in the given system, $\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$, can be written in standard form by subtracting $\frac{7}{4}$ and py from both sides of the equation, which yields $\frac{5}{4}x - py = \frac{8}{4}$. Multiplying each term in this equation by 4 yields $5x - 4py = 8$. The coefficient of x in the first equation, $5x - 14y = -\frac{32}{7}$, is equal to the coefficient of x in the second equation, $5x - 4py = 8$. For the lines to be parallel, and for the coefficients for x and y in one equation to be proportional to the corresponding coefficients in the other equation, the coefficient of y in the second equation must also be equal to the coefficient of y in the first equation. Therefore, $-14 = -4p$. Dividing both sides of this equation by -4 yields $\frac{-14}{-4} = p$, or $p = \frac{7}{2}$. Therefore, if the given system of equations has no solution, the value of p is $\frac{7}{2}$. Note that $7/2$ and 3.5 are examples of ways to enter a correct answer.

Q6**A****A.** 0

Choice A is correct. It is given that line ℓ is perpendicular to a line whose equation is $x = 2$. A line whose equation is a constant value of x is vertical, so ℓ must therefore be horizontal. Horizontal lines have a slope of 0, so ℓ has a slope of 0.

Choice B is incorrect. A line with slope $-\frac{1}{2}$ is perpendicular to a line with slope 2. However, the line with equation $x = 2$ is vertical and has undefined slope (not slope of 2). Choice C is incorrect. A line with slope -2 is perpendicular to a line with slope $\frac{1}{2}$. However, the line with equation $x = 2$ has undefined slope (not slope of $\frac{1}{2}$). Choice D is incorrect; this is the slope of the line $x = 2$ itself, not the slope of a line perpendicular to it.

Q7**B****B.** $hx = 5x + 7$

Choice B is correct. An equation that defines a linear function f can be written in the form $fx = mx + b$, where m and b are constants. It's given in the table that when $x = -4$, $fx = 0$. Substituting -4 for x and 0 for fx in the equation $fx = mx + b$ yields $0 = m(-4) + b$, or $0 = -4m + b$. Adding $4m$ to both sides of this equation yields $4m = b$. Substituting $4m$ for b in the equation $fx = mx + b$ yields $fx = mx + 4m$. It's also given in the table that when $x = -\frac{19}{5}$, $fx = 1$. Substituting $-\frac{19}{5}$ for x and 1 for fx in the equation $fx = mx + 4m$ yields $1 = m(-\frac{19}{5}) + 4m$, or $1 = -\frac{19}{5}m + 4m$. Multiplying both sides of this equation by 5 yields $5 = -19m + 20m$, or $5 = m$. Substituting 5 for m in the equation $fx = mx + 4m$ yields $fx = 5x + 20$. If $hx = fx - 13$, substituting $5x + 20$ for fx in this equation yields $hx = 5x + 20 - 13$, or $hx = 5x + 7$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is an equation that defines the linear function f , not h .

Q8**28**

The correct answer is 28. The minimum number of individual trips for which the cost of the monthly pass is less than the cost of individual tickets can be found by assuming the maximum cost of the individual tickets, \$3.50. If n tickets costing \$3.50 each are purchased in one month, the inequality $95 < 3.50n$ represents this situation. Dividing both sides of the inequality by 3.50 yields $27.14 < n$, which is equivalent to $n > 27.14$. Since only a whole number of tickets can be purchased, it follows that 28 is the minimum number of trips.

Q9**A**

A. Exactly one

Choice A is correct. Distributing 12 on the left-hand side and -3 on the right-hand side of the given equation yields $12x - 36 = -3x - 36$. Adding $3x$ to each side of this equation yields $15x - 36 = -36$. Adding 36 to each side of this equation yields $15x = 0$. Dividing each side of this equation by 15 yields $x = 0$. This means that 0 is the only solution to the given equation. Therefore, the given equation has exactly one solution.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

The correct answer is 8. The given system of equations can be solved using the elimination method. Multiplying both sides of the second equation in the given system by -2 yields $-2ty = -1 - 4x$, or $-1 - 4x = -2ty$. Adding this equation to the first equation in the given system, $4x - 6y = 10y + 2$, yields $4x - 6y + -1 - 4x = 10y + 2 + -2ty$, or $-1 - 6y = 10y - 2ty + 2$. Subtracting $10y$ from both sides of this equation yields $-1 - 6y - 10y = 10y - 2ty + 2 - 10y$, or $-1 - 16y = -2ty + 2$. If the given system has no solution, then the equation $-1 - 16y = -2ty + 2$ has no solution. If this equation has no solution, the coefficients of y on each side of the equation, -16 and $-2t$, must be equal, which yields the equation $-16 = -2t$. Dividing both sides of this equation by -2 yields $8 = t$. Thus, if the system has no solution, the value of t is 8.

Alternate approach: A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are parallel and distinct. Lines represented by equations in the form $Ax + By = C$, where A , B , and C are constant terms, are parallel if the ratio of the x -coefficients is equal to the ratio of the y -coefficients, and distinct if the ratio of the x -coefficients are not equal to the ratio of the constant terms. Subtracting $10y$ from both sides of the first equation in the given system yields $4x - 6y - 10y = 10y + 2 - 10y$, or $4x - 16y = 2$. Subtracting $2x$ from both sides of the second equation in the given system yields $ty - 2x = \frac{1}{2} + 2x - 2x$, or $-2x + ty = \frac{1}{2}$. The ratio of the x -coefficients for these equations is $-\frac{2}{4}$, or $-\frac{1}{2}$. The ratio of the y -coefficients for these equations is $-\frac{t}{16}$. The ratio of the constant terms for these equations is $\frac{\frac{1}{2}}{2}$, or $\frac{1}{4}$. Since the ratio of the x -coefficients, $-\frac{1}{2}$, is not equal to the ratio of the constants, $\frac{1}{4}$, the lines represented by the equations are distinct. Setting the ratio of the x -coefficients equal to the ratio of the y -coefficients yields $-\frac{1}{2} = -\frac{t}{16}$. Multiplying both sides of this equation by -16 yields $-\frac{1}{2} \cdot 16 = -\frac{t}{16} \cdot 16$, or $t = 8$. Therefore, when $t = 8$, the lines represented by these equations are parallel. Thus, if the system has no solution, the value of t is 8.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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